

## **Subject Description Form**

| <b>Subject Code</b>                                   | EE3012 / EE3012B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |   |                               |          |  |  |   |   |   |          |   |   |   |           |   |   |   |                          |   |   |   |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-------------------------------|----------|--|--|---|---|---|----------|---|---|---|-----------|---|---|---|--------------------------|---|---|---|
| <b>Subject Title</b>                                  | Transport Operations Modelling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |   |                               |          |  |  |   |   |   |          |   |   |   |           |   |   |   |                          |   |   |   |
| <b>Credit Value</b>                                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |   |                               |          |  |  |   |   |   |          |   |   |   |           |   |   |   |                          |   |   |   |
| <b>Level</b>                                          | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |   |                               |          |  |  |   |   |   |          |   |   |   |           |   |   |   |                          |   |   |   |
| <b>Pre-requisite/<br/>Co-requisite/<br/>Exclusion</b> | Nil                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |   |                               |          |  |  |   |   |   |          |   |   |   |           |   |   |   |                          |   |   |   |
| <b>Objectives</b>                                     | <ol style="list-style-type: none"> <li>1. To introduce macroscopic and microscopic simulation techniques for transport operations modelling.</li> <li>2. To provide a sound understanding of the theories used in transport operations modelling.</li> <li>3. To enable the building, calibration and validation of transport models.</li> <li>4. To explain the simplifications in modelling and the interpretation of modelling results.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |                               |          |  |  |   |   |   |          |   |   |   |           |   |   |   |                          |   |   |   |
| <b>Intended Learning Outcomes</b>                     | <p>Upon completion of the subject, students will be able to:</p> <ol style="list-style-type: none"> <li>a. Understand the fundamentals and theoretical knowledge of transport modelling and simulation.</li> <li>b. Formulate, apply and assess the transport modelling techniques.</li> <li>c. Understand the strength and limitations of various transport models.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |   |                               |          |  |  |   |   |   |          |   |   |   |           |   |   |   |                          |   |   |   |
| <b>Subject Synopsis/<br/>Indicative Syllabus</b>      | <ul style="list-style-type: none"> <li>• Introduction to transport operations modelling (macro and micro)</li> <li>• Car following and lane changing models – Gipps, IDM, MOBIL, etc.</li> <li>• Use of microscopic simulation software (SUMO, Aimsun or Vissim)</li> <li>• Macroscopic dynamic traffic flow model (Cell Transmission Model, CTM)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |                               |          |  |  |   |   |   |          |   |   |   |           |   |   |   |                          |   |   |   |
| <b>Teaching/Learning Methodology</b>                  | <p>Delivery of the subject is mainly through formal lectures and complemented by tutorials. Assignments and projects provide students hands-on experience in modelling, while report-writing enables students to practise writing skill.</p> <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th rowspan="2">Teaching/Learning Methodology</th><th colspan="3">Outcomes</th></tr> <tr> <th>a</th><th>b</th><th>c</th></tr> </thead> <tbody> <tr> <td>Lectures</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td></tr> <tr> <td>Tutorials</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td></tr> <tr> <td>Assignments and projects</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td></tr> </tbody> </table> |   |   | Teaching/Learning Methodology | Outcomes |  |  | a | b | c | Lectures | ✓ | ✓ | ✓ | Tutorials | ✓ | ✓ | ✓ | Assignments and projects | ✓ | ✓ | ✓ |
| Teaching/Learning Methodology                         | Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |   |                               |          |  |  |   |   |   |          |   |   |   |           |   |   |   |                          |   |   |   |
|                                                       | a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | b | c |                               |          |  |  |   |   |   |          |   |   |   |           |   |   |   |                          |   |   |   |
| Lectures                                              | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | ✓ | ✓ |                               |          |  |  |   |   |   |          |   |   |   |           |   |   |   |                          |   |   |   |
| Tutorials                                             | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | ✓ | ✓ |                               |          |  |  |   |   |   |          |   |   |   |           |   |   |   |                          |   |   |   |
| Assignments and projects                              | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | ✓ | ✓ |                               |          |  |  |   |   |   |          |   |   |   |           |   |   |   |                          |   |   |   |

|                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                                                   |   |          |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------------------|---|----------|
| <b>Assessment Methods in Alignment with Intended Learning Outcomes</b> | Specific assessment methods/tasks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | % weighting | Intended subject learning outcomes to be assessed |   |          |
|                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             | a                                                 | b | c        |
|                                                                        | 1. Written examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 45%         | ✓                                                 | ✓ | ✓        |
|                                                                        | 2. Assignments and exercises                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 30%         | ✓                                                 | ✓ | ✓        |
|                                                                        | 3. Projects                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 25%         |                                                   | ✓ | ✓        |
|                                                                        | Total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 100%        |                                                   |   |          |
|                                                                        | Examination and test allow assessment on outcomes covering principles, techniques, design and application. Assignments and projects enable students to explore and apply transport modelling techniques and evaluate transport operations performance.                                                                                                                                                                                                                                                                                    |             |                                                   |   |          |
| <b>Student Study Effort Expected</b>                                   | Class contact:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                                   |   |          |
|                                                                        | ▪ Lecture / Tutorial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |                                                   |   | 39 Hrs.  |
|                                                                        | Other student study effort:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                                   |   |          |
|                                                                        | ▪ Assignments and projects                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |                                                   |   | 35 Hrs.  |
|                                                                        | ▪ Self-study                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |                                                   |   | 33 Hrs.  |
|                                                                        | Total student study effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |                                                   |   | 107 Hrs. |
| <b>Reading List and References</b>                                     | <ul style="list-style-type: none"><li>Ni, D. Traffic Flow Theory: Characteristics, Experimental Methods, and Numerical Techniques, Elsevier, 2015.</li><li>Hoogendoorn S.P., Knoop V. Traffic Flow Theory and Simulation. <a href="https://ocw.tudelft.nl/courses/traffic-flow-theory-simulation/">https://ocw.tudelft.nl/courses/traffic-flow-theory-simulation/</a> Accessed on 23 July 2025.</li><li>Banks, J., Carson, J.S., Nelson, B.L., Nicol, D.M. Discrete-Event System Simulation (3rd Edition). Prentice Hall, 2000.</li></ul> |             |                                                   |   |          |